



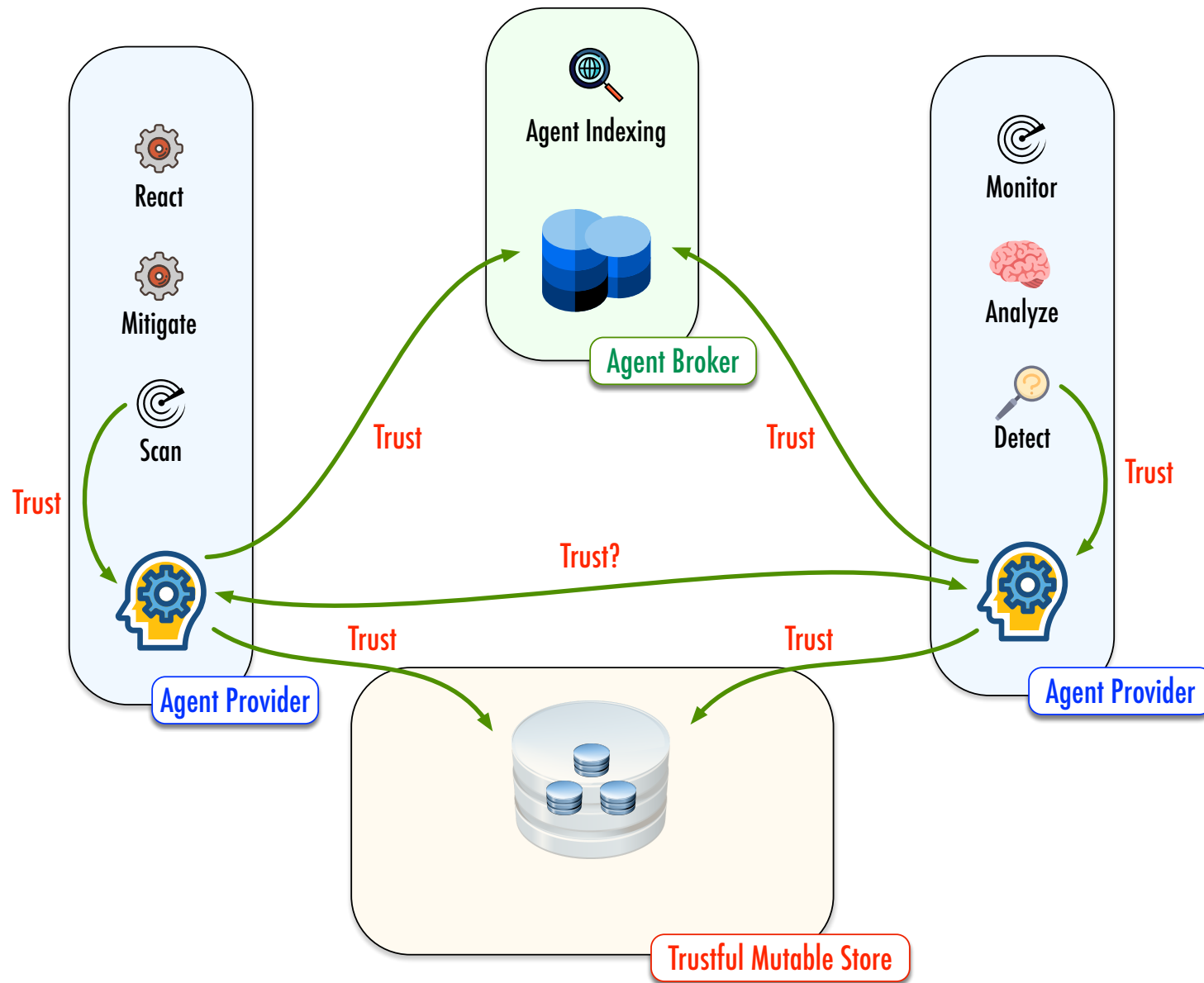
Self-Certifying Identity and Capability-Based Delegation for Autonomous AI Agents

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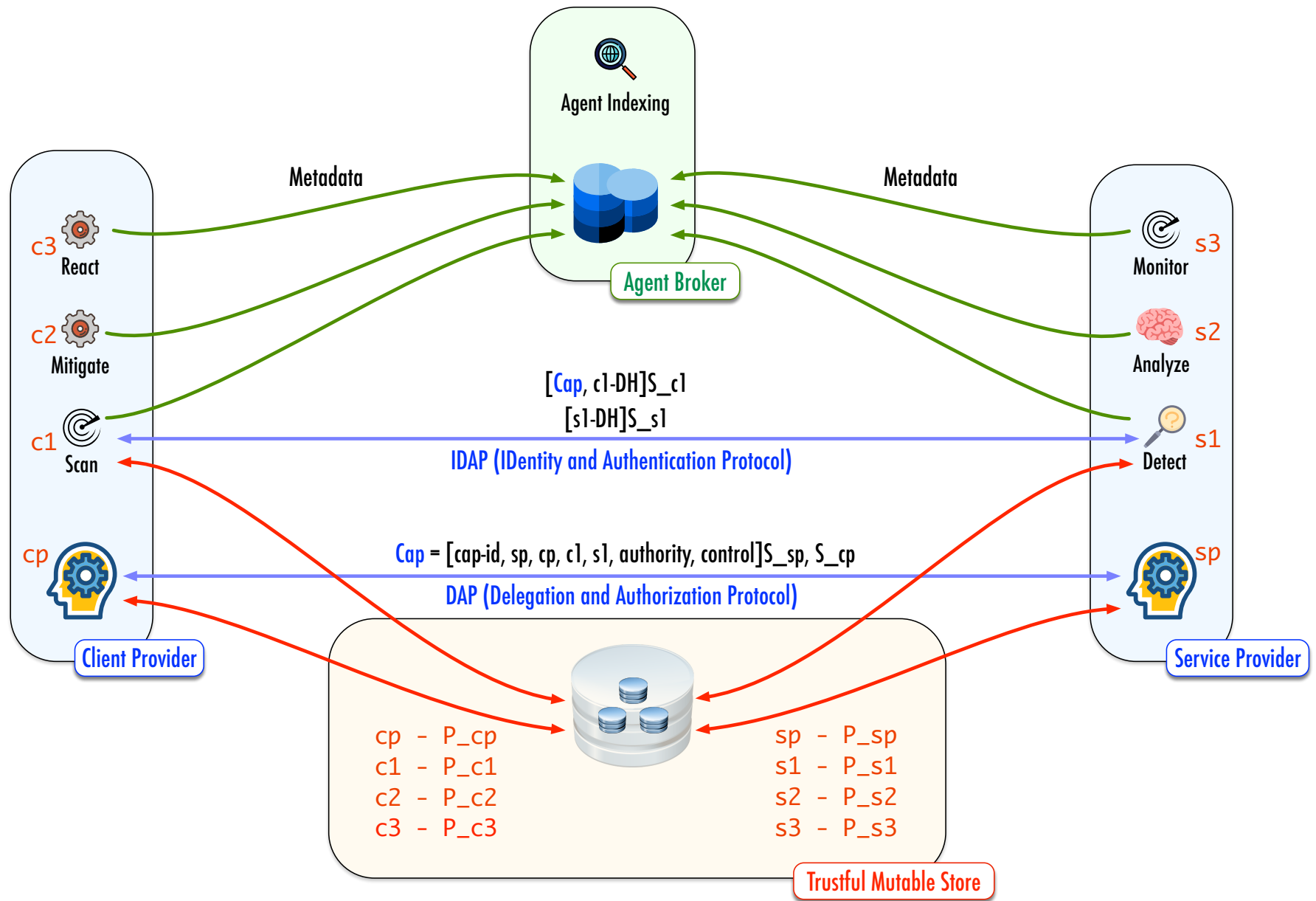
Huawei Technologies France



SID - Self-certifying IDentifiers

- Public key P and secret key S
- Hash of public key P similarly to Bitcoin addresses:

$$\text{SID} = \text{Context} \mid \text{CI}$$
$$\text{CI} = \text{ripemd160}(\text{sha256}(P))$$
- SID is encoded with **base32** (32 characters), e.g.,
 1exq5yqb16148pf0fu7juh1tjkrbu2rxf
- SID can be a **domain name**
- Agent can prove the **possession of the SID** based on **the private key**
- Need trustful **binding** between SID and public key



DAP and IDAP

- Client Provider negotiates with Service Provider
 - **capability** that defines the entities involved in the mission, conditions and limits of the mission, protocol requirements
 - DAP (Delegation and Authorization Protocol)
- Client AI Agent
 - has **identity** and uses **capability** for task execution by a Service AI Agent
 - IDAP (IDentity and Authentication Protocol)

DNS as Trustful Mutable Store

- Agent descriptions at `_agents.openai.com`.
- An agent has a domain name:
`1exq5yqbl6l48pf0fu7juhl tjkrbu2rxf.openai.com`.
- `1exq5yqbl6l48pf0fu7juhl tjkrbu2rxf TXT`
 `"sid=1exq5yqbl6l48pf0fu7juhl tjkrbu2rxf";`
 `"key=1caac9c64711f66e6ed71b37dc...";`
 `"proto="MCP";`
 `"name="image-analyzer";`
 `"desc="Analyzes images and provides text`
 `descriptions";`
 `"in-type="image/png,image/jpeg";`
 `"out-type="text/plain,application/json";`
 `"meta="http://openai.com/agents/1exq5yqbl6l48pf0f`
 `u7juhl tjkrbu2rxf";`

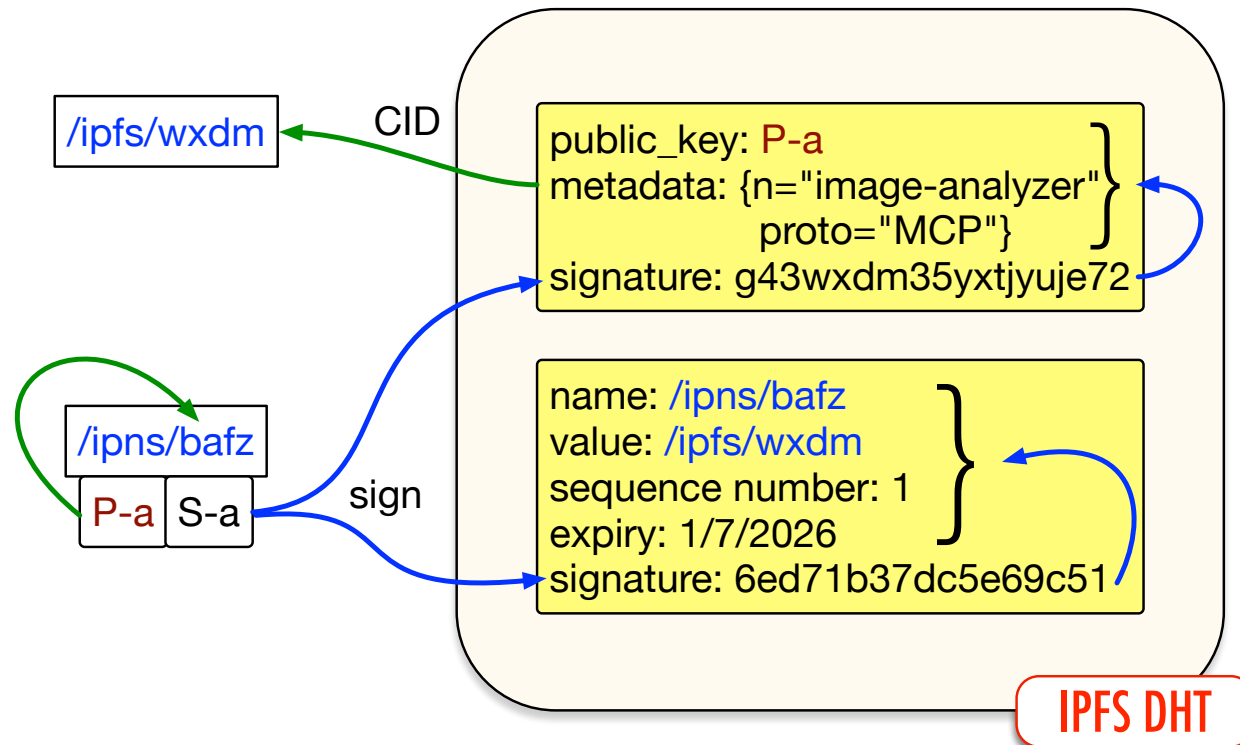
IPFS as Trustful Mutable Store

- InterPlanetary File System (IPFS)
 - hash(content) → CID (Content Identifier)
 - network of Peers that store content
 - Distributed Hash Table (DHT) maps CIDs to Peers
 - immutable, distributed, highly available
- InterPlanetary Naming System (IPNS)
 - hash (public key) → IPNS name
 - IPNS record stored in DHT:
 - IPNS name
 - IPFS CID to immutable metadata
 - public key, validity, purpose, binding signature
 - sequence number, expiry, signature by private key

IID - IPFS IDentifiers

- Public key P and secret key S
- IID - IPFS Identifier: hash of public key P
$$\text{IID} = \text{base32}(\text{hash}(P))$$
- $P = 1caac9c64711f66e6ed\dots \rightarrow$
 /ipns/bafz...
- Metadata object signed with S :
 /ipfs/wxdm...
- $\{\text{public_key: } 1caac9c64711f66e6ed\dots,$
- $\text{ metadata: \{n="image-analyzer"$
- $\text{ proto="MCP"\},$
- $\text{ signature: g43wxdm35yxtjyuje72\dots}\}$

IID – Public Key Binding



- **P-a** – public key → `/ipns/bafz` → `/ipfs/wxdm`
 - IID and metadata object
- **S-a** – secret key for signatures

Conclusion

- Identity = Identifier + Key → SID or IID
 - derived from public keys, another form of DID
- Trustful Mutable Store
 - DNS/DNSSEC or IPFS
- Trust = Name + Identifier + Key
 - DNS as a unifying namespace
 - trust linked to entities – AI Agent Providers or Agent Brokers
 - trust based on cryptographic continuity
- Protocols
 - DAP for negotiating Capability between trusted entities
 - IDAP for authentication/authorization based on Capability

IID - IPFS IDentifiers

- Public key P and secret key S
- IID - IPFS Identifier: hash of public key P
IID = base32(hash(P))
- $P = 1caac9c64711f66e6ed\dots \rightarrow$
/ipns/bafzbeia...
- Metadata object signed with S :
/ipfs/metadata-CID
- {public_key: 1caac9c64711f66e6ed...,
- metadata: {n="image-analyzer"
- proto="MCP"},
- signature: g43wxdm35yxtjyuje72...}

IID – Public Key Binding

- IPNS IID
- `/ipns/bafzbeia...`
- Metadata in IPFS
- `/ipfs/metadata-CID`
- Their binding in IPNS record stored in DHT:
 - `name: /ipns/bafzbeia...`
 - `value: /ipfs/metadata-CID`
 - `sequence number: 1`
 - `expiry: 1/7/2026`
 - `signature: 6ed71b37dc5e69c51...`
- Signed with *S*

IID – Trust Binding

- Metadata object signed with *S*:
/ipfs/metadata-CID
- {*public_key*: 1caac9c64711f66e6ed...,
- *metadata*: {*n*="image-analyzer"
- *proto*="MCP"} ,
- *signature*: g43wxdm35yxtjyuje72...
- }
- Metadata object can be signed by another entity, e.g., *Agent Provider*

IID – Trust Binding

- Publish in a DNSSEC-protected domain
- TXT record at `bafzbeia.openai.com`.
- Agent IID:
- `bafzbeia` TXT
 - `"iid=bafzbeia";`
 - `"key=1caac9c64711f66e6ed...";`
- Trust: `Agent Provider` → `domain` → `IID`
- Binding between `AI Agent Provider` and `IID`
- IPFS object also provides `metadata` for indexing